

# O2 Data Efficiency Report

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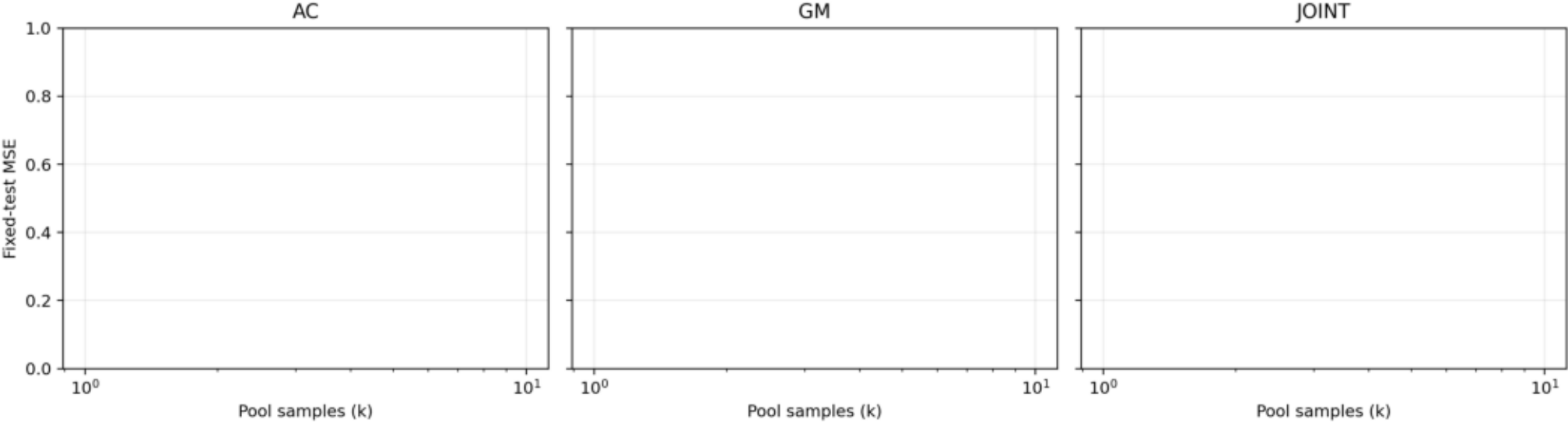
Fixed evaluation set: 474 held-out GMs x 57 ACs = 27,018 samples.  
Reduced supervision varies GM/source diversity and AC/intensity scales.  
Models: FNO-Large, Transformer, BiLSTM. E-Base imported from O1.

- FNO-Large: E-Base MSE=0.0449, R2=0.9423; reduced runs pending.

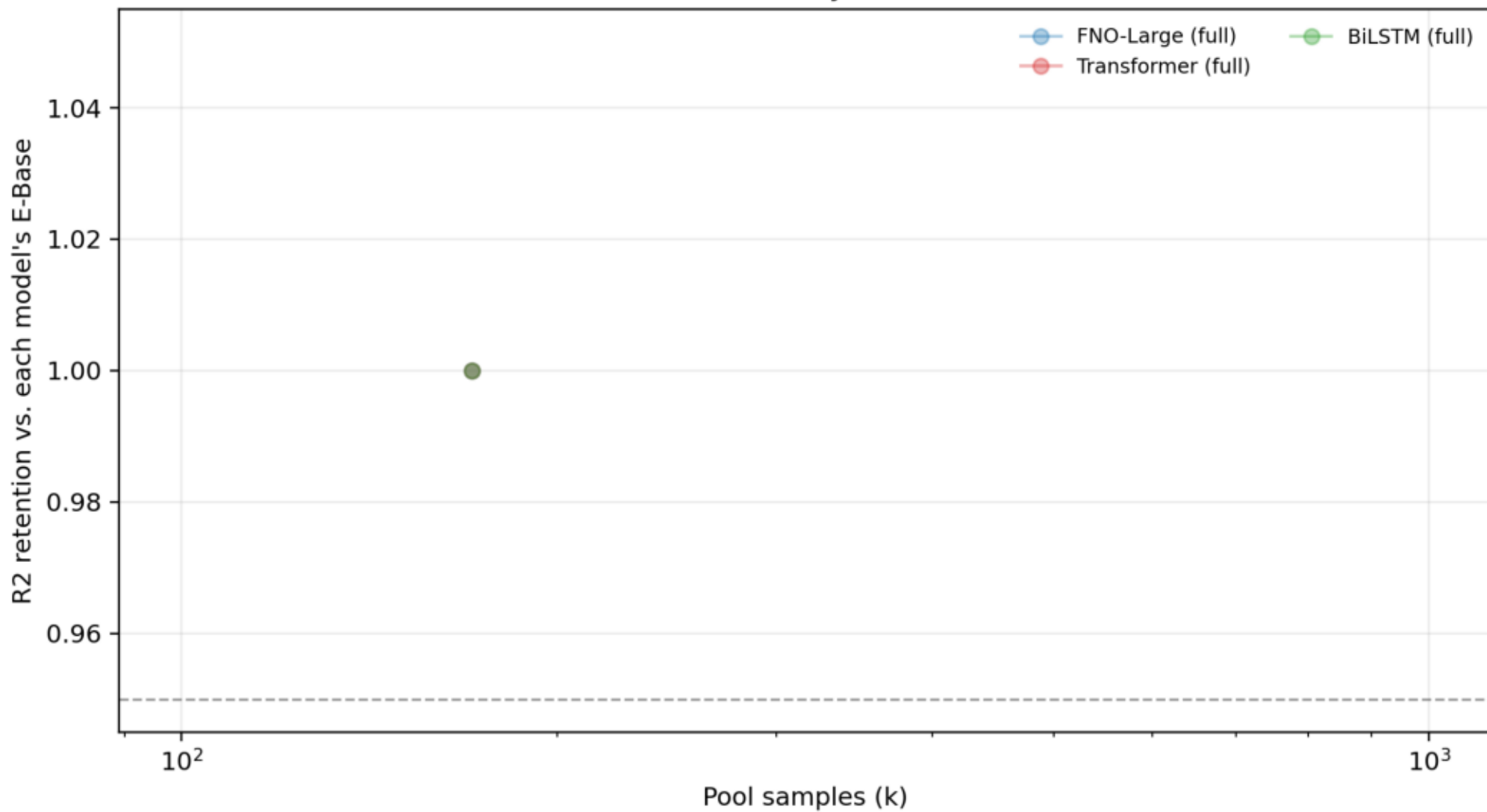
- Transformer: E-Base MSE=0.0214, R2=0.9725; reduced runs pending.

- BiLSTM: E-Base MSE=0.0099, R2=0.9872; reduced runs pending.

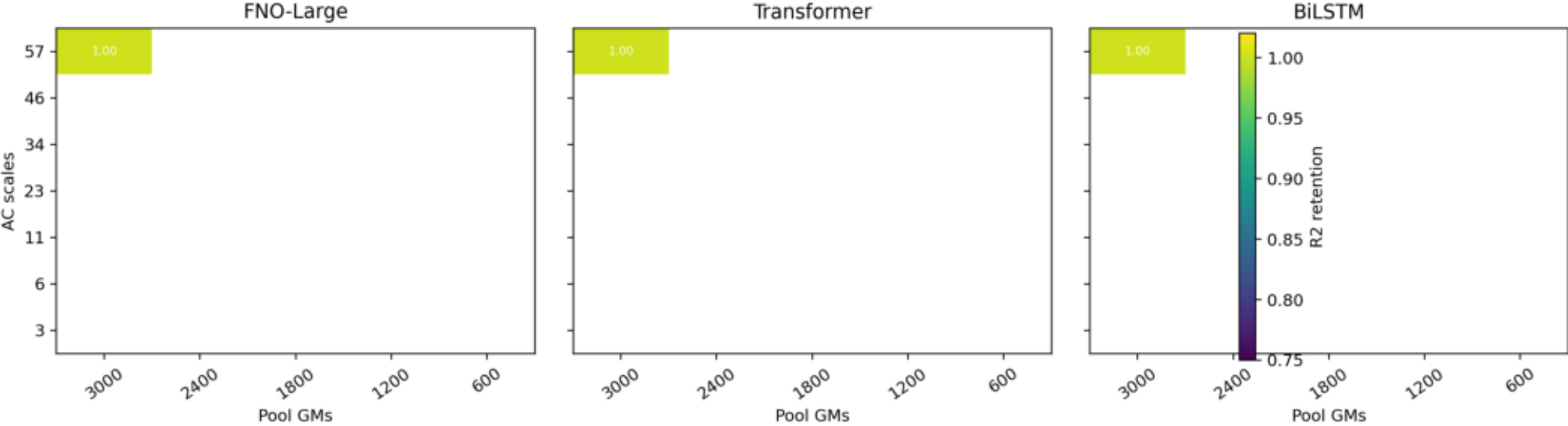
O2 Data Efficiency: Test MSE Degradation



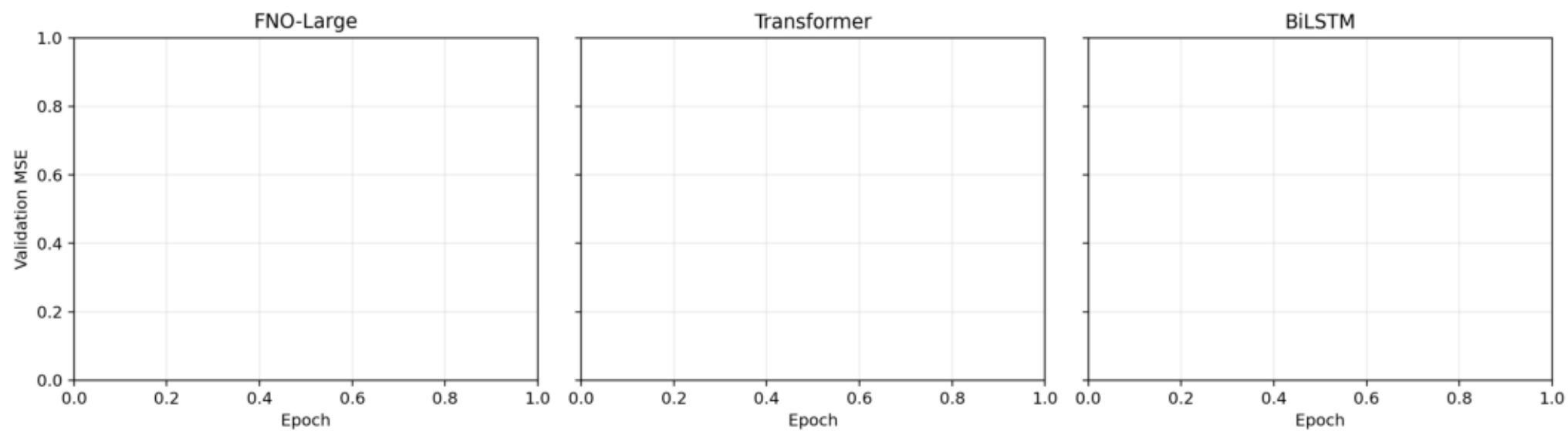
Relative Fidelity Retention



Retention Heatmap over GM and AC Budgets



Representative Reduced-Data Validation Curves



| Model       | Run    | GMs  | ACs | Pool   | MSE    | R2     | R2 Ret. |
|-------------|--------|------|-----|--------|--------|--------|---------|
| FNO-Large   | E-Base | 3000 | 57  | 171.0k | 0.0449 | 0.9423 | 1.0000  |
| Transformer | E-Base | 3000 | 57  | 171.0k | 0.0214 | 0.9725 | 1.0000  |
| BiLSTM      | E-Base | 3000 | 57  | 171.0k | 0.0099 | 0.9872 | 1.0000  |